



GENE THERAPY PROCESS MANUFACTURING MAPS

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About BioPhorum

BioPhorum's mission is to create environments where the global biopharmaceutical industry can collaborate and accelerate its rate of progress, for the benefit of all.

Since its inception in 2004, BioPhorum has become the open and trusted environment where senior leaders of the biopharmaceutical industry come together to openly share and discuss the emerging trends and challenges facing their industry.

Growing from an end-user group in 2008, BioPhorum now comprises over 90 manufacturers and suppliers deploying their top 3,500 leaders and subject matter experts to work in seven focused Phorums, articulating the industry's technology roadmap, defining the supply partner practices of the future, and developing and adopting best practices in drug substance, fill finish, process development and manufacturing IT. In each of these Phorums, BioPhorum facilitators bring leaders together to create future visions, mobilize teams of experts on the opportunities, create partnerships that enable change and provide the quickest route to implementation, so that the industry shares, learns and builds the best solutions together.

1.0

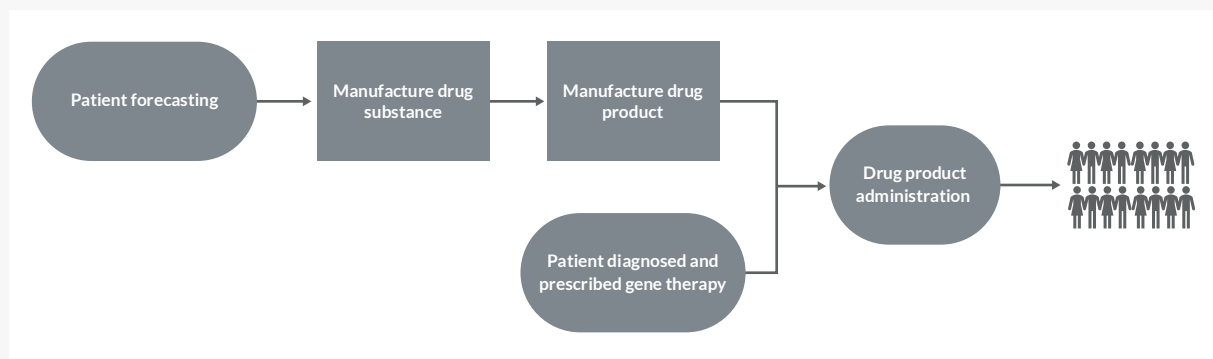
Introduction

Gene therapy has undergone rapid growth over the last few years. The validity of this platform as an effective treatment for various disease indications has been established through recent commercial approvals for both *in vivo* (e.g. gene delivery through viral vectors) and *ex vivo* (e.g. gene delivery through genetically modified cells such as chimeric antigen receptor therapy (CAR-T)) gene therapies (also known as gene-modified cell therapy). Platforms for manufacturing of viral vectors (as final product or initial starting materials) for these therapies are still evolving, with constant improvements being made to processes, resulting in increased yields and quality of the product. As companies are working in many different therapeutic areas, and are employing a wide range of processes and procedures, BioPhorum's Cell & Gene Therapy Commercialization workstream has generated a set of gene therapy process maps that represent example processes, unit operations and platform technologies being used in *in vivo* gene therapy manufacturing (viral vectors).

A separate publication is being created by the BioPhorum Cell & Gene Therapy Commercialization workstream covering gene-modified cell therapy / *ex vivo* gene therapy www.biophorum.com/download/cell-therapy-process-map/

The gene therapy process maps provided in the appendices are intended to be used as an initial introduction to the *in vivo* gene therapy manufacture processes and so act as an educational tool for people new to the field and form a general platform baseline. Additionally, the baseline maps can be used to help drive discussions and identify specific areas that require further expansion. A high-level overview of *in vivo* gene therapy is shown in Figure 1.

Figure 1: *in vivo* gene therapy manufacture overview



In vivo gene therapy is the direct administration of a gene of interest to genetically complement a defective endogenous gene and restore biological function in a patient; or encode for the production of a therapeutic protein. The most common system used to deliver the gene of interest *in vivo* is a viral vector such as adeno-associated virus (AAV). On the other hand, in *ex vivo* gene therapy, cells are extracted from a patient or healthy donor, transduced with the gene of interest using a gene-delivery vector such as lentivirus vectors (LVV). The transduced cells, which are now armed with disease-fighting capability such as killing tumors, are then infused back to the patient. Therefore, viral vectors such as AAVs and LVVs have become an important tool for gene therapies and their efficient manufacturing has become a central topic to make commercialization of these lifesaving treatments viable.

AAVs are non-pathogenic and replication defective and have emerged as the predominant vector for *in vivo* gene therapy. AAVs are small (25 nm diameter), non-enveloped viruses belonging to the Parvoviridae family. There are numerous different AAV serotypes. The various AAV serotypes differ in their capsid components and so display differential tissue tropism, transduction efficiency and immunogenicity. The AAV genome is packaged in a capsid, containing three structural proteins, VP1, VP2, and VP3,

that assemble in 60-subunit icosahedral geometry to encapsulate the viral single-stranded DNA genome of ~4.8 kilobase (kb). Production and formulation of AAV products requires carefully optimized processing conditions to ensure proper encapsulation and assembly of the viral particles, minimize AAV titer loss during bioprocessing, and prevent viral particle aggregation. Long-term shelf life requires storage at very low temperatures of <-60°C.

LVVs are single-stranded RNA vectors derived from the human immunodeficiency virus that have emerged as the predominant vector used in *ex vivo* gene therapies. LVVs can infect different cell types including non-dividing primary cells with high transducing efficiency and induce prolonged, stable gene expression within the target cell. LVVs belong to the family Retroviridae, are enveloped, and 80–120 nm in diameter. LVVs can accommodate transgenes in the range of 9–10 kb, thereby overcoming a major limitation of AAVs (i.e. packaging capacity of ~4.8 kb transgene). LVVs are used in the transduction of T cells in the manufacturing process of CAR-T cell therapy. Production and formulation of LVVs requires optimized bioprocess manufacturing process to ensure viral titer. The size and the lipid bilayer envelope of LVVs makes sterile filtration through a 0.2 µm filter challenging due to viral titer loss in the filtration process. Like AAV, long-term shelf life requires storage at low temperatures of <-60°C.

2.0

In/out of scope

The gene therapy maps presented here are not intended to be comprehensive in scope. Guidance is included on items in scope for these maps as well as on those that are out of scope. The details included in the gene therapy maps (e.g. specific cell lines, unit operations) are not meant to be prescriptive in nature, but to represent specific industry-relevant examples for such workflows, with the understanding that alternate approaches continue to be developed.

In scope

- model workflows describing manufacturing of adeno-associated virus (AAV) and lentivirus (LVV) viral vectors for use in *in vivo* or *ex vivo* gene therapy applications
- workflows represent manufacturing approaches that are most commonly in use across the industry
- workflows for initial starting materials that require cell culture / fermentation steps for manufacturing, e.g. plasmid DNA and cell banks
- brief explanation of example technologies for each process step box (not an exhaustive list).

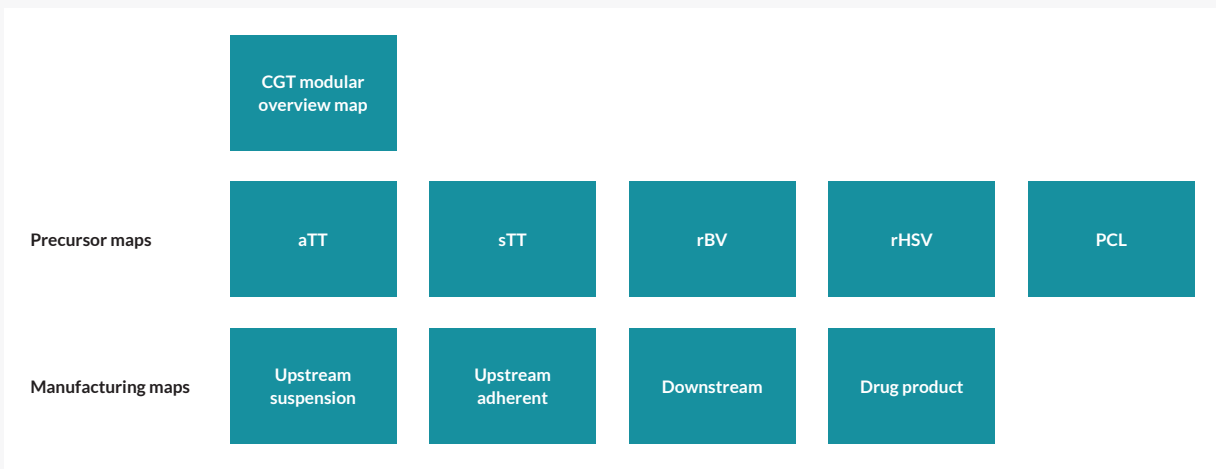
Out of scope

- exhaustive list of all processes, methods, and technologies (including any emerging technologies) for gene therapy manufacturing
- quality control sampling points – study is focused on manufacturing aspects and assume each company will test at required points as per own control strategy and regulatory requirements
- pros/cons of technology platforms
- detailed step-by-step methodology indicating each equipment/handling step
- timing – processing or turnaround.

3.0

Overview of the process maps

Each box is a hyperlink to the full process map, with legend, in the appendices.



Key:

aTT = adherent Transient Transfection; sTT = suspension Transient Transfection; rBV = recombinant baculovirus; rHSV = recombinant herpes simplex virus; PCL = producer cell line

The process maps in the appendices are not intended as a full description of the process, rather they are intended to show unit operation steps representative of a typical manufacture process. Each process flow and unit operation should be reviewed on a case-by-case basis, and each step can be considered 'optional', depending on requirement.

Each map contains steps above the workflow describing points for addition of buffers, raw materials, and key ancillary materials.

4.0

Conclusions

Viral vector manufacturing is a relatively nascent field. Currently, the industry relies on a diverse set of production platforms. The process maps describe a series of production platforms that require a unique set of precursor materials to support drug substance manufacturing. The approach for cell bank production is consistent across all production platforms, a two-tier cell bank system is manufactured to support life cycle management of the target program. The two-tier banking system is extended to the process viruses (rHSV, rBV and Ad5). Plasmid DNA is manufactured using a similar approach with a master cell bank (MCB) which is then used to produce plasmid DNA for drug substance manufacturing. Although the precursor materials may appear quite different, it should be noted the unit operations and production methods are consistent across all platforms. For viral vector drug substance manufacturing, the approach is consistent across all platforms – culture cells, initiate viral vector production using the precursor material, and then purify using a series of chromatography columns and filtration devices.

The goal of this collaboration was to provide visual maps, summarizing the inputs and outputs and processing steps of viral vector manufacturing. These maps provide a lens to identify and acknowledge the complexity of the starting materials required for several manufacturing platforms. However, this complex approach is also executed using traditional biologics manufacturing unit operations: bioreactors, liquid chromatography columns, and tangential-flow filtration membranes. Users and audiences can leverage the maps to identify potential areas for future process development and application of advanced manufacturing technologies. The maps were created by a team of subject matter experts from 13 different organizations and companies. It is hoped that audiences of novices and experts can leverage the maps to enhance communication, facilitate knowledge transfer and dissemination of current strategies for viral vector manufacturing.

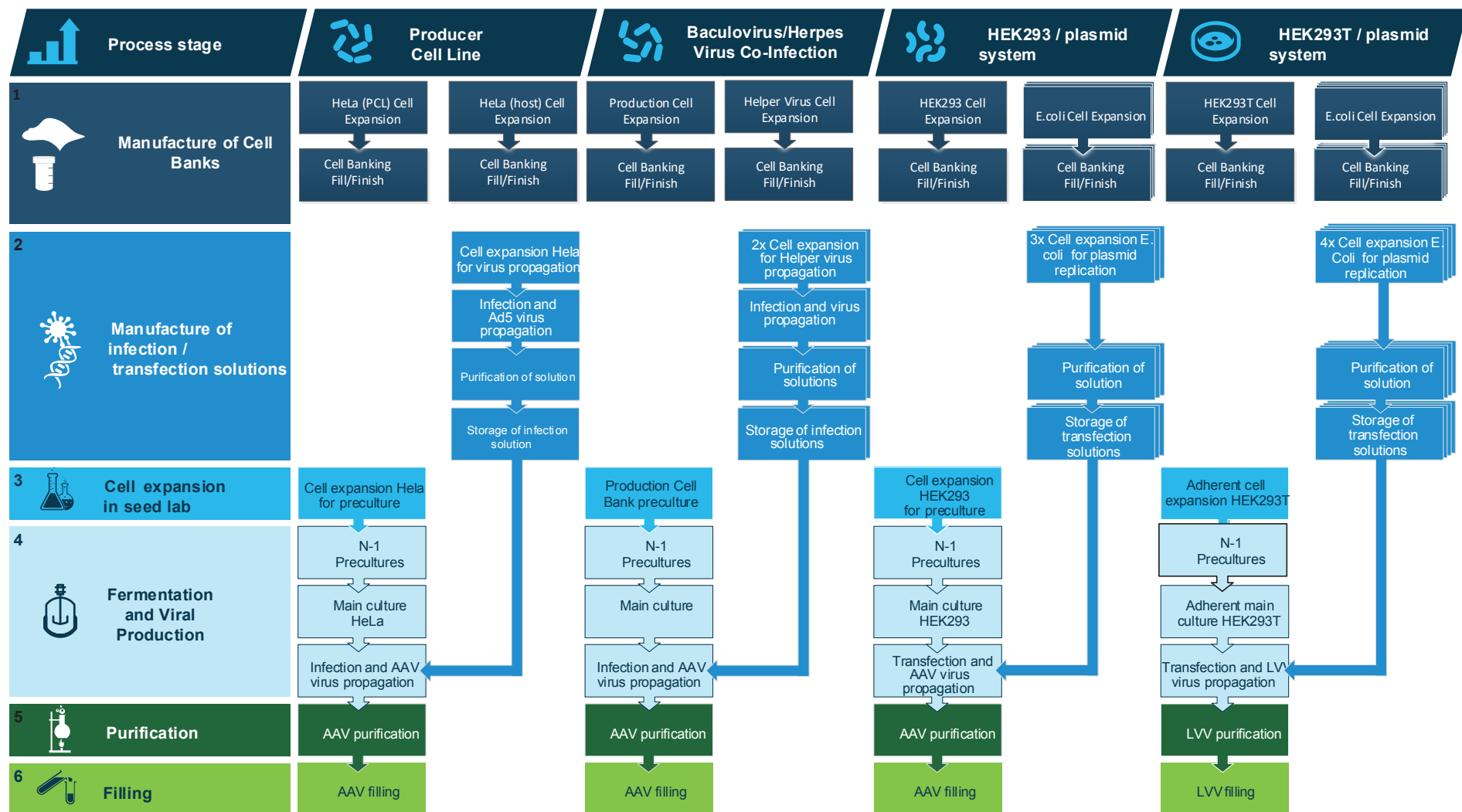
Appendices

Process maps

The following ten-page appendices contain the process maps.

These process maps are also available to download and print in A3.

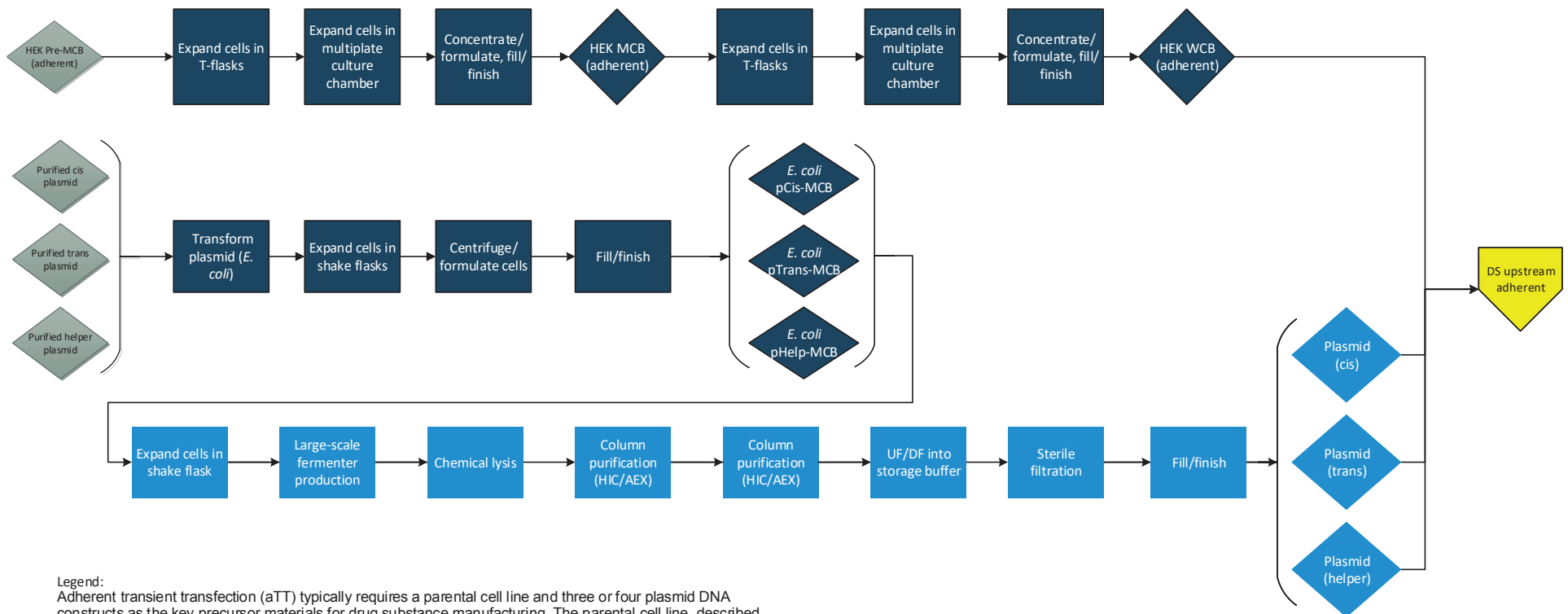
Appendix 1: Modular overview map



LEGEND:
There are five primary viral vector manufacturing platforms described in this exercise. Four describe AAV production platforms: a producer cell line, recombinant baculovirus co-infection of Sf9 cells (rBV), Herpes Simplex Virus co-infection of BHK cells (rHSV), a triple plasmid DNA transfection of HEK293 cells. A fifth platform describes a four plasmid DNA transfection of HEK293 cells for the production of LVV.

Note – all maps presented in the appendix show unit operation steps that are intended only to be representative of a typical manufacture process. Therefore, each process flow should be reviewed on a case by case basis, and every unit operation can be viewed as “Optional”.

Appendix 2: Precursor aTT

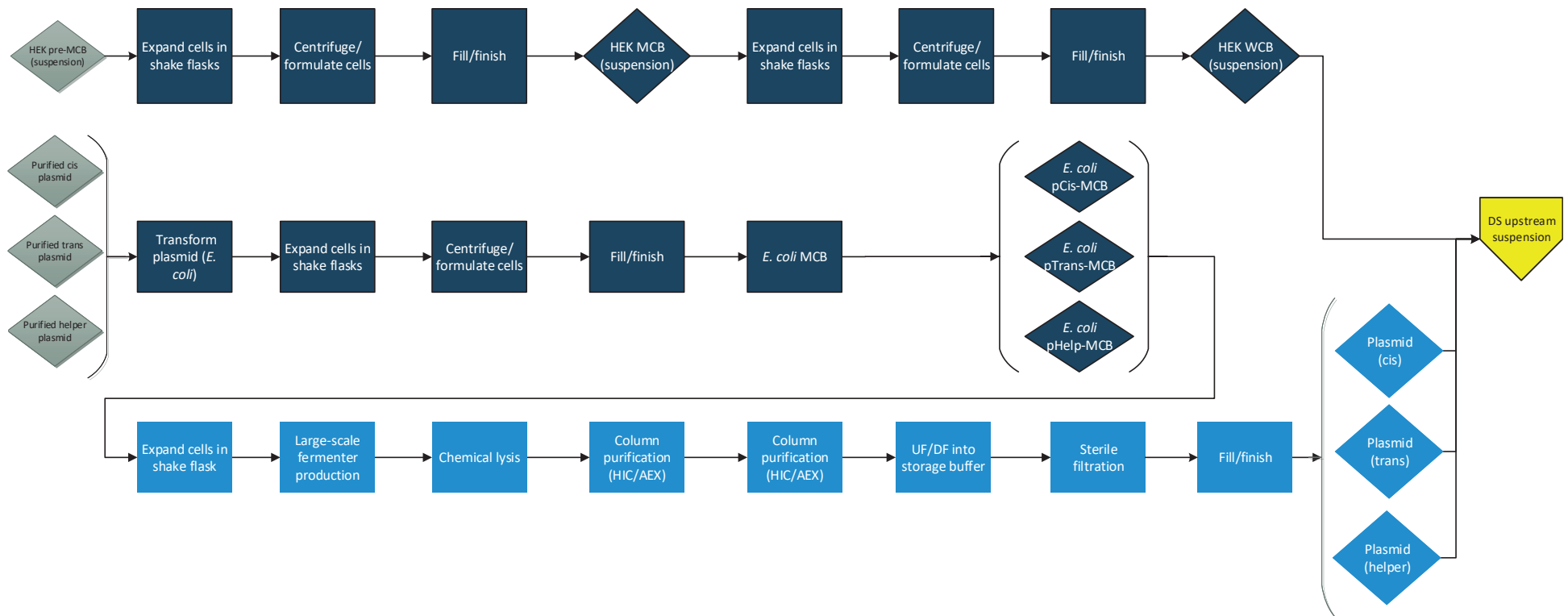


Legend:

Adherent transient transfection (aTT) typically requires a parental cell line and three or four plasmid DNA constructs as the key precursor materials for drug substance manufacturing. The parental cell line, described here as an adherent human embryonic kidney (HEK), is cultured directly on growth surfaces. The cells are serially expanded first to create a master cell bank (MCB) and second to create a working cell bank (WCB). The plasmid DNA constructs are transformed and amplified using *E. coli* in shake flasks to first produce an MCB. The MCB is then cultured in a large-scale fermenter and cells are lysed to release the plasmid DNA. After several column-based purification methods, the purified DNA is stored at -60°C .

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Appendix 3: Precursor sTT

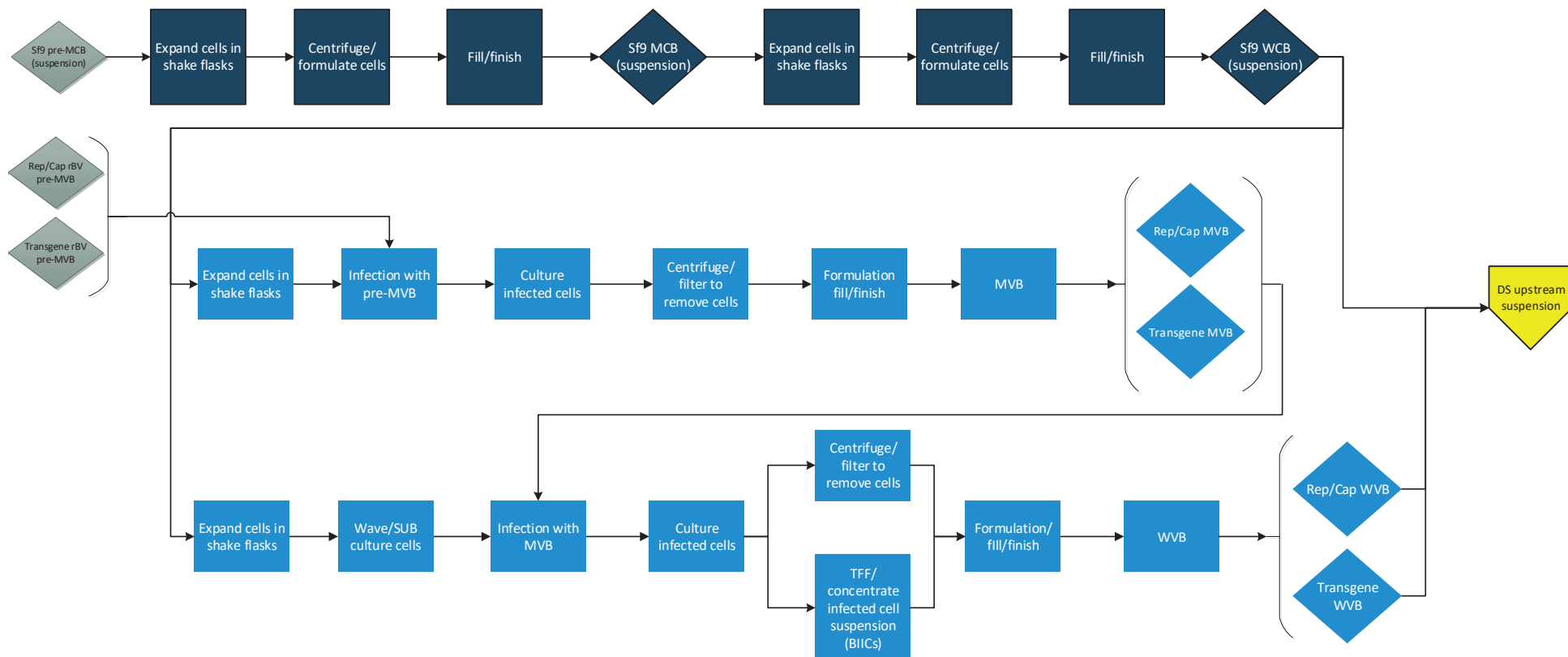


Legend:

Suspension transient transfection (sTT) production platforms typically require a parental cell line and three or four plasmid DNA constructs as the key precursor materials for drug substance manufacturing. The parental cell line described here as a suspension adapted HEK cell line, is cultured in shake flasks. The cells are serially expanded first to create an MCB and second to create a WCB. The plasmids described in the adherent transient transfection map can be used as a precursor material for this platform.

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Appendix 4: Precursor rBV

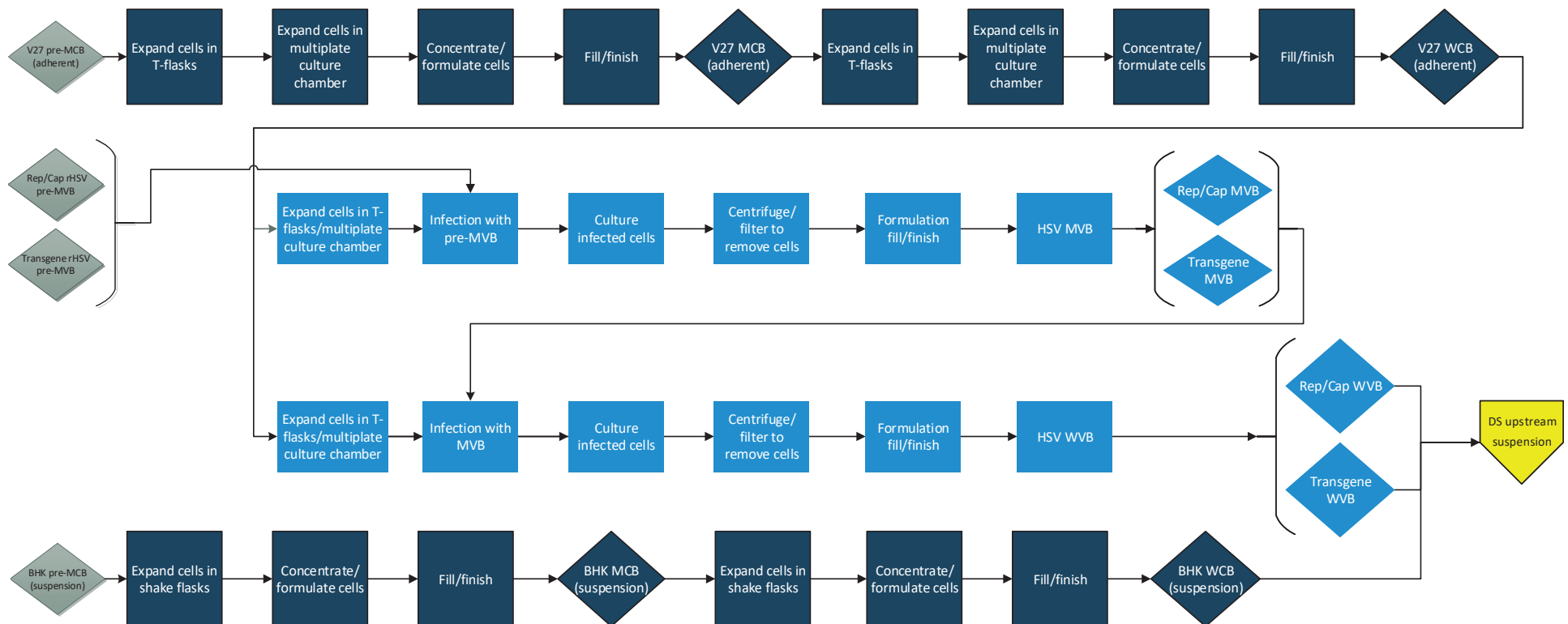


Legend:

Recombinant baculovirus (rBV) production platforms typically require a parental cell line and two recombinant baculovirus constructs as the key precursor materials for drug substance manufacturing. The parental cell line described here as a suspension adapted *Spodoptera frugiperda* (Sf9) is cultured in shake flasks. The cells are serially expanded first to create an MCB and second to create a WCB. The WCB is used to amplify the baculovirus constructs in shake flasks and purified by removal of cells/cell debris to prepare a rBV MVB. Next, the WCB is used to amplify the rBV MVB in a larger volume, using a rocking or stirred-tank bioreactor. The culture is purified by removal of cells/cell debris and an optional concentration step to prepare the rBV WVB.

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Appendix 5: Precursor rHSV

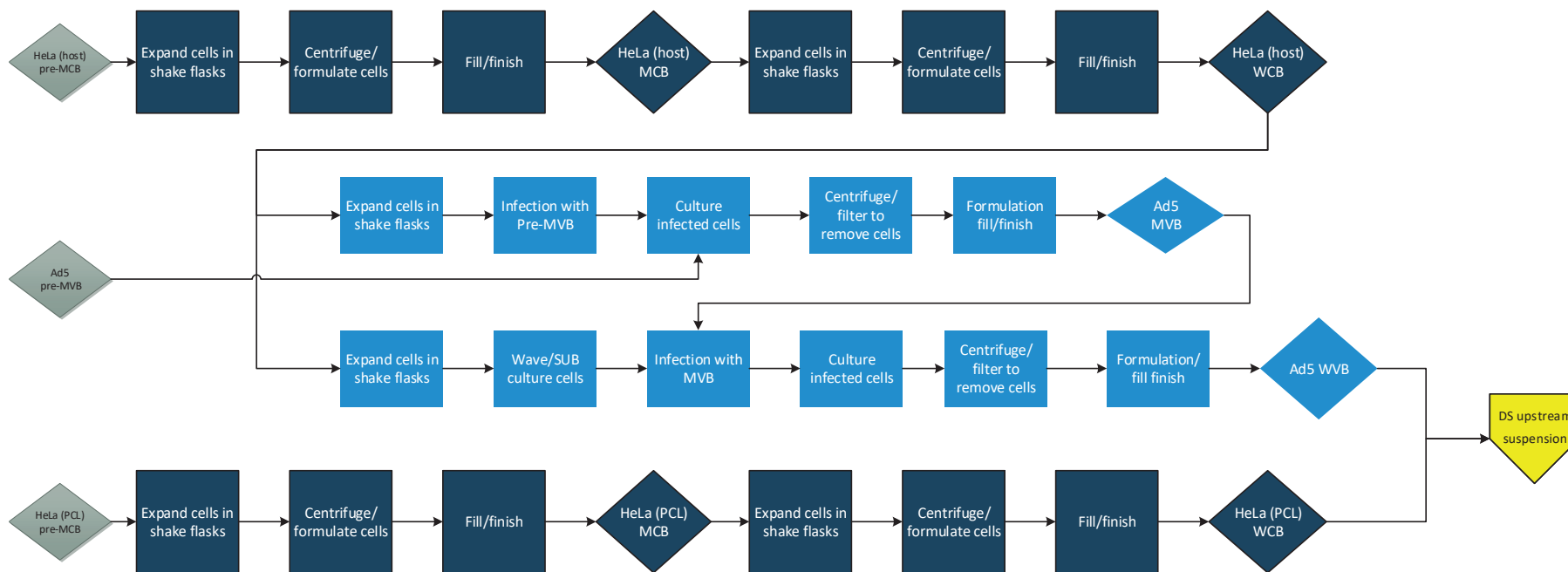


Legend:

Recombinant herpes simplex virus (rHSV) production platforms typically require two cell lines and two recombinant rHSV constructs as the key precursor materials for drug substance manufacturing. The first cell line is an adherent, recombinant vero cell line that contains a stably transfected copy of the ICP27 gene. The cells are cultured directly on growth surfaces to create an MCB and then a WCB. The WCB is used to amplify the rHSV constructs (with a deletion of the ICP27 gene) and purified by removal of the cells/cell debris to prepare an rHSV MVB. Next, the WCB is used to amplify the rHSV MVB using a larger surface area. The culture is purified by the removal of cells/cell debris and a concentration step to prepare the rHSV WVB. The second cell line is a suspension-adapted parental BHK cell line cultured in shake flasks that serves as the production host cell line for AAV manufacture. The cells are serially expanded first to create an MCB and second to create a WCB.

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Appendix 6: Precursor PCL

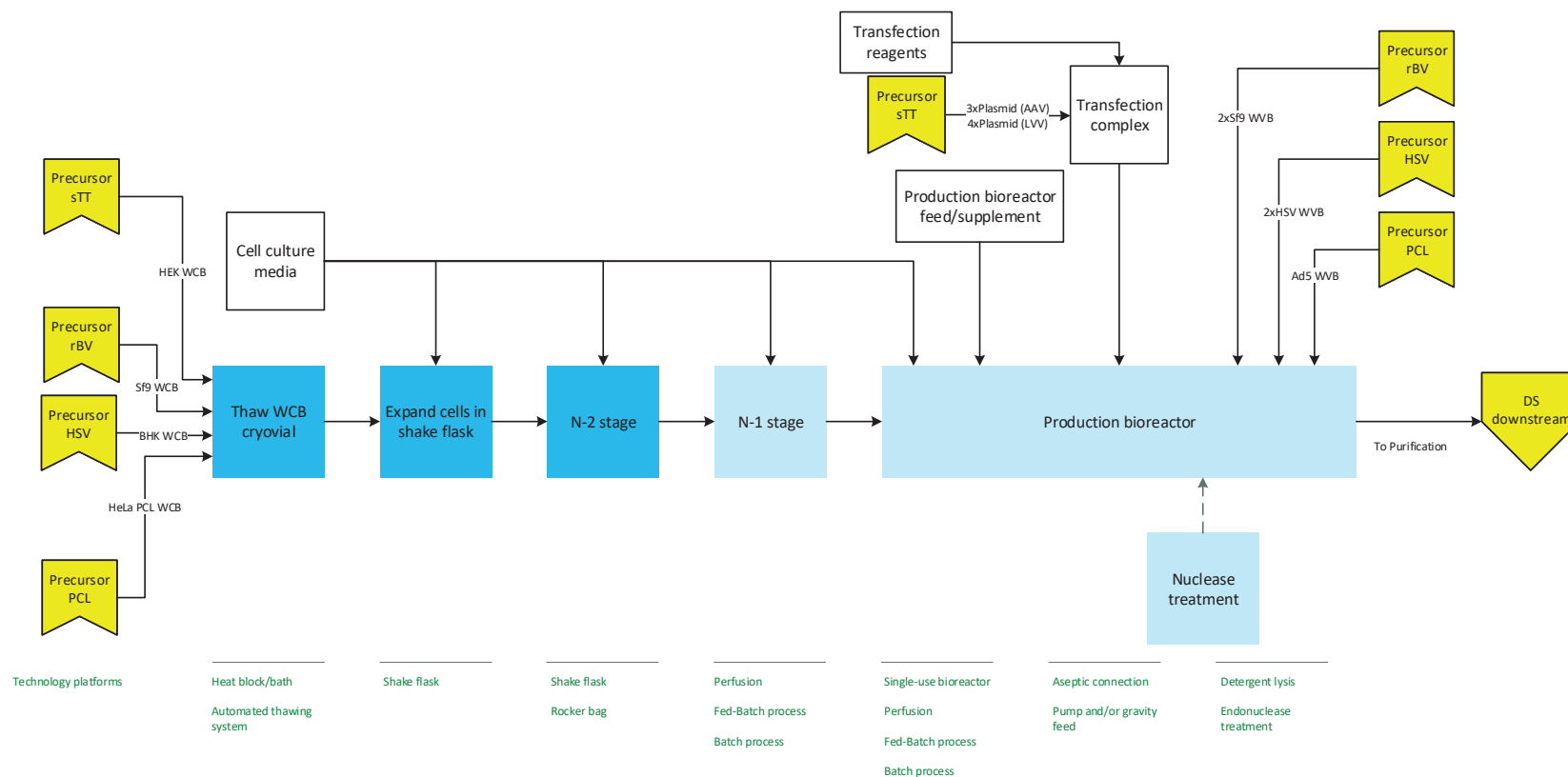


Legend:

Producer cell line (PCL) production platforms will typically require two cell lines and a helper Adenovirus (Ad5) as the key precursor materials for drug substance manufacturing. The first cell line is a suspension-adapted parental HeLa cell line cultured in shake flasks. The cells are serially expanded first to create an MCB and second to create a WCB. The WCB is used to amplify the Ad5 virus and purified by removal of the cells/cell debris to prepare an MVB. The WCB is used to amplify Ad5 in shake flasks and purified by removal of cells/cell debris to prepare an Ad5 MVB. Next, the WCB is used to amplify the Ad5 MVB in a larger volume, using a rocking or stirred-tank bioreactor. The culture is purified by removal of cells/cell debris to prepare the Ad5 WVB. The second cell line is a suspension-adapted recombinant HeLa cell line containing a stably transfected transgene and rep/cap genes that serve as the production host cell line for AAV manufacture. In shake flasks, the cells are serially expanded first to create an MCB and second to create a WCB.

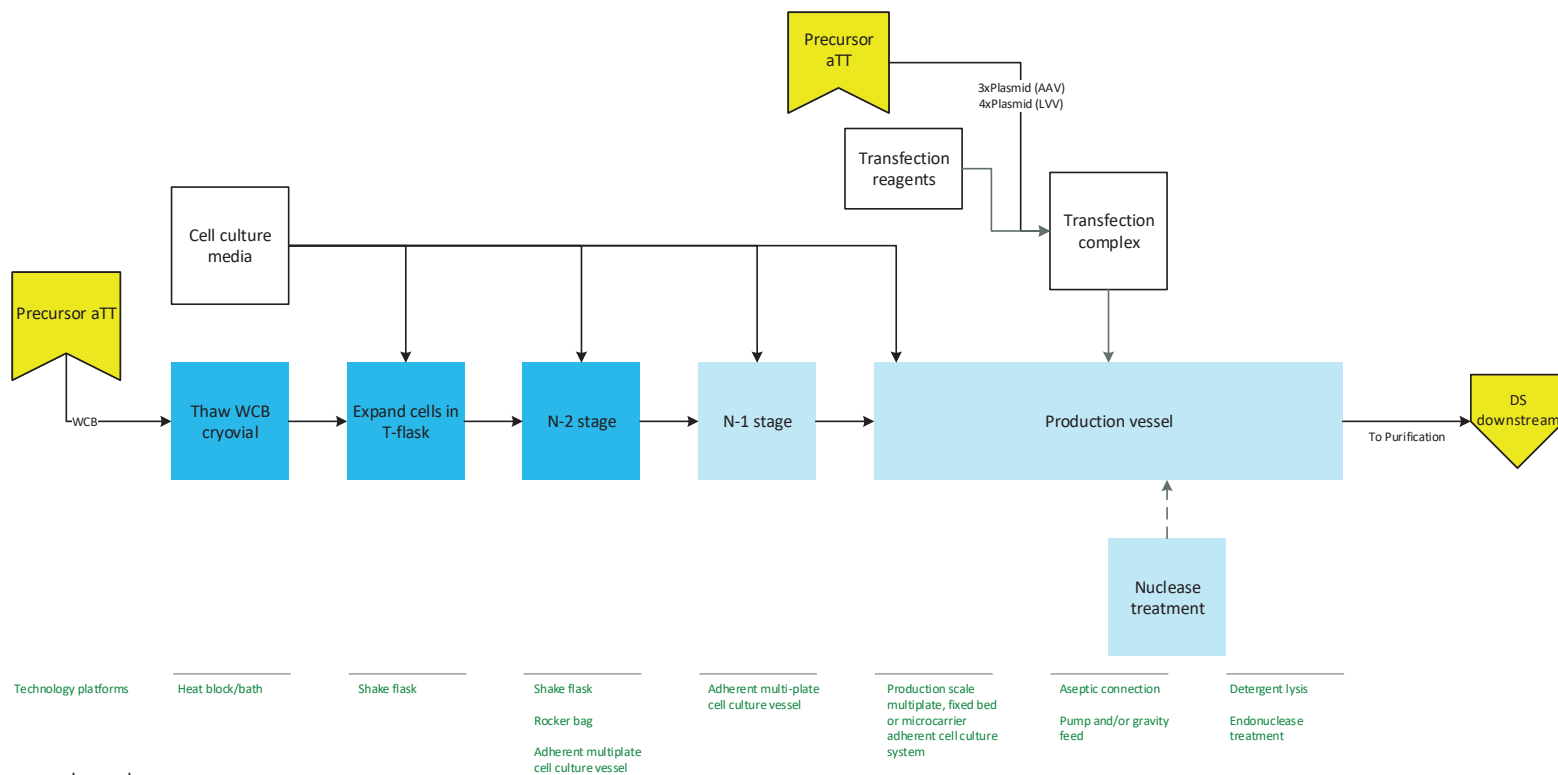
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Appendix 7: Manufacture drug substance upstream (suspension)



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Appendix 8: Manufacture drug substance upstream (adherent)

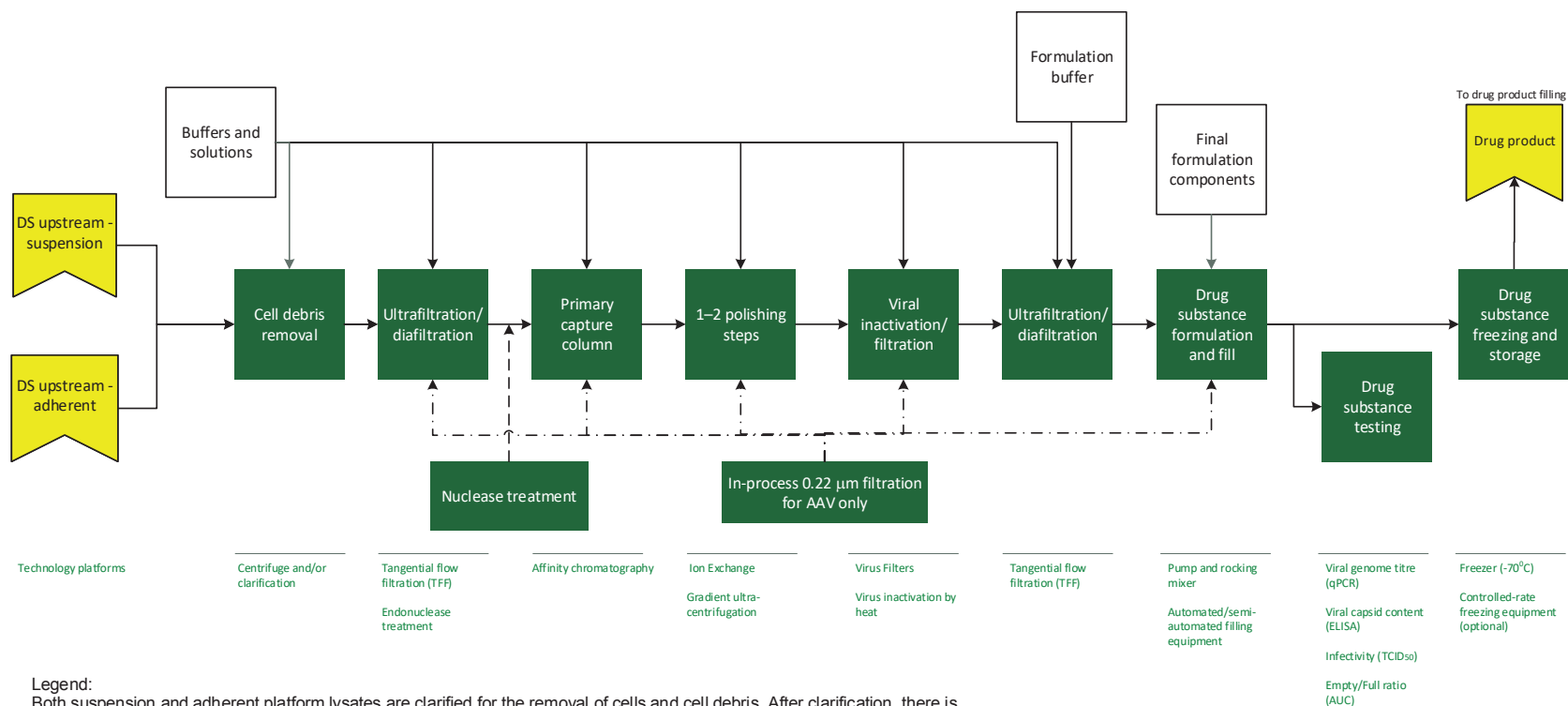


Legend:

The adherent-based platforms will typically expand the WCB using T-flasks and multi-layer culture flasks to seed either a production bioreactor or a series of multi-layer culture flasks. After a predefined cell culture that could require a feed or supplement, viral vector production is initiated by the addition of a platform-specific precursor material - plasmid DNA. After the viral vector production phase, the harvest is initiated with the optional addition of a detergent for cell lysis and/or an endonuclease for degradation of host cell nucleic acids, residual process DNA, or incomplete assembly of AAV particles.

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Appendix 9: Manufacture drug substance downstream

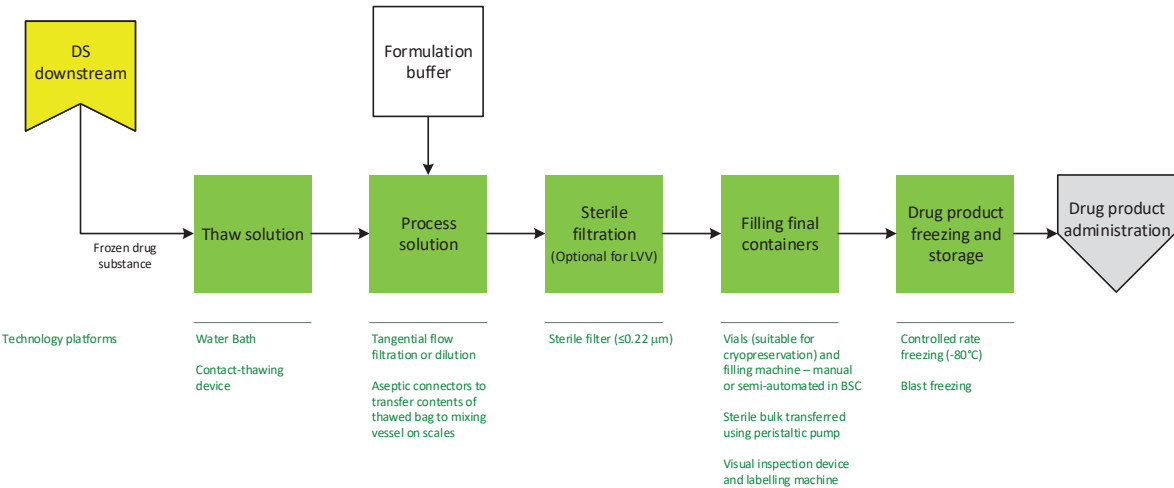


Legend:

Both suspension and adherent platform lysates are clarified for the removal of cells and cell debris. After clarification, there is an optional concentration and buffer exchange step and endonuclease treatment. A primary capture column designed to target the viral vector is used for the removal of process and product impurities. For AAV processes, a 1-2 column polishing step could be required to reduce empty particles and to further remove process and product impurities. For LVV processes, a column-polishing step may not be necessary. The purified viral vector solution could be concentrated and formulated to prepare for drug substance bottling. To control bioburden in the downstream process, a 0.22 µm filter can be employed between the unit operations.

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Appendix 10: Manufacture drug product



Legend:
The thawed drug substance can be formulated to target a specific viral vector concentration and sterile filtered prior to fill and finish. The final drug product is typically frozen at < -60°C.
However, it should be noted that for LVV used in *ex vivo* gene therapy this would be part of the 'drug substance' process and freeze/thaw is not required.

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